# Real Analysis HW 1

1. (Ch 2. 6) Let $A$ be the set of irrational numbers in the interval $[0,1]$, Prove that $m^*(A)=1$

Pf. 依无理数定义 $[0,1]\setminus A = \mathbb{Q}\cap[0,1]$，由可数集零测

$$m^*([0,1]\setminus A) = m^*(\mathbb{Q}\cap[0,1]) = 0$$

由可数次可加性，$[0,1] = ([0,1]\setminus A)\cup A$，有

$$m^*([0,1]) \le m^*([0,1]\setminus A) + m^*(A) = m^*(A)$$

由单调性

$$m^*(A) \le m^*([0,1])$$

故

$$m^*(A) = m^*([0,1]) = 1$$

□

2. (Ch 2. 9) Prove that if $m^*(A)=0$, then $m^*(A\cup B) = m^*(B)$

Pf. 由单调性

$$m^*(B) \le m^*(A\cup B)$$

由可数次可加性

$$m^*(A\cup B) \le m^*(A) + m^*(B) = m^*(B)$$

从而

$$m^*(A\cup B) = m^*(B)$$

□

3. (Ch 2. 14) Show that if a set $E$ has positive outer measure then there is a bounded subset of $E$ that also has positive outer measure

Pf. 反证. 考虑

$$\mathbb{R} = \bigcup_{t\in\mathbb{Z}} [t,t+1] \Rightarrow E = \bigcup_{t\in\mathbb{Z}} [t,t+1]\cap E$$

由于$\mathbb{Z}$与$\mathbb{N}$等势，可把 $[t,t+1]\cap E$ $(t\in\mathbb{Z})$ 排列为 $\{I_k\}_{k=1}^{\infty}$. 则每个 $I_k$ 有界，由假设零测，由可数次可加性有矛盾

$$0 < m^*(E) = m^*\left(\bigcup_{k=1}^{\infty} I_k\right) \le \sum_{k=1}^{\infty} m^*(I_k) = 0$$

□

4. (Ch 2. 15) Show that if $E$ has finite measure and $\varepsilon > 0$, then $E$ is the disjoint union of a finite number of measurable sets, each of which has measure at most $\varepsilon$.

Pf. 考虑 $[t\varepsilon, (t+1)\varepsilon)$ 是 $\cap E$ 是可测集之交，从而可测.

$$\mathbb{R} = \bigcup_{t\in\mathbb{Z}} [t\varepsilon, (t+1)\varepsilon) \Longrightarrow E = \bigcup_{t\in\mathbb{Z}} [t\varepsilon, (t+1)\varepsilon) \cap E$$

将 $[t\varepsilon, (t+1)\varepsilon) \cap E$ $(t\in\mathbb{Z})$ 排列为 $\{I_k\}_{k=1}^{\infty}$，则 $E = \bigcup_{k=1}^{\infty} I_k$，由单调性

$$m^*(I_k) \le m^*([t\varepsilon, (t+1)\varepsilon)) = \varepsilon$$

注意到对任意 $n\in\mathbb{N}$，由于 $I_k$ 可测，有

$$m^*(E) = \sum_{k=1}^{n} m^*(I_k) + m^*\left(\bigcup_{k=n+1}^{\infty} I_k\right)$$

由单调性

$$m^*(E) = m^*\left(\bigcup_{k=1}^{\infty} I_k\right) \ge m^*\left(\bigcup_{k=1}^{n} I_k\right) = \sum_{k=1}^{n} m^*(I_k)$$

由可数次可加性

$$m^*(E) = m^*\left(\bigcup_{k=1}^{\infty} I_k\right) \le \sum_{k=1}^{\infty} m^*(I_k).$$

故

$$m^*(E) = \sum_{k=1}^{\infty} m^*(I_k) < \infty$$

由极限的性质存在 $n_0\in\mathbb{N}$，使得

$$\sum_{k=n_0+1}^{\infty} m^*(I_k) \le \varepsilon$$

进

$$m^*\left(\bigcup_{k=n_0+1}^{\infty} I_k\right) \le \sum_{k=n_0+1}^{\infty} m^*(I_k) \le \varepsilon$$

这表明

$$E = \left(\bigcup_{k=1}^{n_0} I_k\right) \cup \left(\bigcup_{k=n_0+1}^{\infty} I_k\right)$$

中 $m^*(I_k) \le \varepsilon$ $(1\le k\le n_0)$，$m^*\left(\bigcup_{k=n_0+1}^{\infty} I_k\right) \le \varepsilon$.

□.

5. (Ch 2. 21) Use property (ii) of Theorem 11 as the primitive definition of a measurable set and prove that the union of two measurable sets is measurable. Then do the same for property (iv).

Pf. (ii) 任取可测集 $E_1, E_2$，则存在 $G_\delta$ 集 $G_1, G_2$ 使得

$$E_1 \subseteq G_1 \quad E_2 \subseteq G_2 \quad m^*(G_1 \setminus E_1) = m^*(G_2 \setminus E_2) = 0$$

考虑 $G = G_1 \cup G_2$，由于 $G_1, G_2$ 是 $G_\delta$ 集，可写为开集的可数交

$$G_1 = \bigcap_{k=1}^{\infty} G_{1k} \qquad G_2 = \bigcap_{k=1}^{\infty} G_{2k}.$$

取

$$G_{11}' = G_{11} \qquad G_{1k}' = \bigcap_{j=1}^{k} G_{1k}$$

则 $G_{1,k+1}' \subset G_{1k}'$. 对 $G_2$ 同样操作，有

$$G = G_1 \cup G_2 = \left(\bigcap_{k=1}^{\infty} G_{1k}'\right) \cup \left(\bigcap_{k=1}^{\infty} G_{2k}'\right) = \bigcap_{k=1}^{\infty} (G_{1k}' \cup G_{2k}')$$

是 $G_\delta$ 集. 注意到

$$\begin{aligned} G \setminus (E_1 \cup E_2) &\subseteq G_1 \setminus ((E_1 \cup E_2) \cap G_1) \cup G_2 \setminus ((E_1 \cup E_2) \cap G_2) \\ &\subseteq (G_1 \setminus E_1) \cup (G_2 \setminus E_2) \end{aligned}$$

$$\Longrightarrow m^*(G \setminus (E_1 \cup E_2)) \leq m^*(G_1 \setminus E_1) + m^*(G_2 \setminus E_2) = 0$$

知 $m^*(G \setminus (E_1 \cup E_2)) = 0$，即 $E_1 \cup E_2$ 可测.

(iv) 同样取 $F_1 \subseteq E_1$ $F_2 \subseteq E_2$，是 $F_\sigma$ 集，使得

$$m^*(E_1 \setminus F_1) = m^*(E_2 \setminus F_2) = 0$$

取可数个闭集并

$$F_1 = \bigcup_{k=1}^{\infty} F_{1k} \qquad F_2 = \bigcup_{k=1}^{\infty} F_{2k}$$

同理可设 $F_{1k} \subset F_{1,k+1}$ $F_{2k} \subseteq F_{2,k+1}$，此及证明

$$(E_1 \cup E_2) \setminus F \subset (E_1 \setminus F_1) \cup (E_2 \setminus F_2)$$

故 $F$ 是 $F_\sigma$ 集且由单调性

$$m^*((E_1 \cup E_2) \setminus F) \leq m^*(E_1 \setminus F_1) + m^*(E_2 \setminus F_2) = 0.$$

$\square$.